CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in agentic systems and applications built on large language models, from tool calling, agent orchestration and MCP servers to RAG and systematic evaluation. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in machine learning and Python development.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present | Muldestausee, Germany

- Sole developer owning an agentic knowledge-extraction system end-to-end, from data ingestion and chunking to prompt engineering and indexing (agentic, deterministic, and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized a multi-agent system with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for regulatory and technical documents, with systematic evaluation against an SAP-system-derived ground truth; drove precision/recall from 33%/45% to 93%/90%.
- Designed and implemented a Model Context Protocol (MCP) server exposing the knowledge graph for millisecond-latency retrieval, integrated into Claude Code; used Claude Code as standard practice with custom agent skills and agentic coding workflows, grounded in clean code principles.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- **Tech stack: Python, LangChain, LangGraph, LiteLLM, LLM APIs, MCP, RAG, GraphRAG, Vector Search, Docker, Kubernetes, Azure, FastAPI, React, Neo4j, MongoDB, Postgres**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 | Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, advised on where agentic AI could support them, and translated the requirements into a solution approach.
- Built and deployed an agentic AI application with LangGraph that routes requests across vector database queries, general inference, and summarization, with a React frontend and a Python FastAPI backend.
- Containerized the stack with Docker and deployed it to AWS with CDK through GitLab CI/CD.
- **Tech stack: Python, LangChain, LangGraph, FastAPI, React, Docker, GitLab CI/CD, AWS (CDK, EC2, Lambda)**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 | Leverkusen, Germany

- Automated evaluation of answer quality and retrieval for the PrismaChat RAG system with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Built retrieval pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for better retrieval.
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, Azure AI Search, AWS CDK**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data, engineering lag features and rolling statistical windows from sensor data.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Spec-Driven Coding Agent Harness | Claude Code | FastMCP | Langfuse | CI/CD

- Harness that turns a plain-language feature request into a testable specification (acceptance criteria written as pytest tests) and lets a coding agent (Claude Code) implement it under a rule set: project conventions, custom skills, and hooks that block edits to protected files and secrets; an MCP server built with FastMCP gives the agent the spec and task context.
- A CI/CD delivery pipeline gates every agent-written change with automated tests, static security analysis, and dependency scanning, plus an evaluation suite over benchmark tasks (pass rate, cost, turns per task); tracing and logging of every agent run go to Langfuse for LLMOps monitoring and observability.

Project: Meme Generation using LLMs | Open Llama 3B | QLoRA

- Fine-tuned the open-source Open Llama 3B model with QLoRA, which accelerated training, to generate responses to educational meme-related prompts.

TECHNICAL SKILLSET

- **Agentic Systems:** agents and multi-agent systems, tool calling, agent orchestration (LangGraph), MCP servers, agent skills (Claude Code), prompt engineering, LLM APIs (LiteLLM), LangChain
- **RAG & Evaluation:** Retrieval-Augmented Generation, GraphRAG, embedding models, RAGAS, PytrecEval, ground-truth evaluation, Hugging Face Transformers, fine-tuning (QLoRA)
- **AI Development Tools:** Claude Code (custom skills, agentic coding workflows, MCP integration)
- **Programming:** Python, SQL, FastAPI, REST APIs, Python microservices, React, Streamlit
- **Vector Databases & Data:** Qdrant, Pinecone, Azure AI Search, Postgres, MongoDB, Neo4j, DynamoDB
- **Delivery Pipelines & Cloud:** Docker, Kubernetes, GitLab CI/CD, Git, AWS (CDK, Lambda, EC2, SQS, S3), Azure
- **Machine Learning:** Scikit-Learn, XGBoost, PyTorch, TensorFlow, time-series analysis

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)